Week	Topics	Focus Area	Python Tasks	Sub - Tasks	Recognition Activities	Assign Faculty
Month 1: Python Programming Foundations						
Week 1	Python Basics: I/O + Data Types	- Intro to Python - Variables: int, float, str, bool - input(), print(), type conversion	- Calculator for 2 numbers - User info collector	Understand data types, accept user input	Students create a tech meme related to Python basics (data types, loops, functions) using Canva or Google Drawings.	**Shivalika Ma'am**
	Conditional Logic + Loops	- if, elif, else - while, for, range() - break, continue	- ATM Simulator (PIN check + actions) - FizzBuzz Game - Mini Project 1: Number Guessing Game	Build logic with loops + decision making		
Week 2	Functions + Lists & Tuples	- def, return, parameters - Lists: slicing, methods - Tuples: immutability, uses	- List of marks: avg, min, max - Factorial using function	Flowchart design of function logic		
	Dict + Set + String Ops	- Dictionary: CRUD - Sets: union, intersection - Strings: .lower(), .replace()	- Contact Book (dictionary-based) - Word Counter in string - Mini Project 2: Report Card Generator	Format output + logical data handling		
Month 2: OOP Concepts + Python Practice						
Week 3	Classes & Objects	- class, __init__, self, attributes/methods	- Car class with speed & brand - Accelerate method	Ask ChatGPT for an OOP analogy + explain class	Groups propose a startup idea where Python OOP can solve a real-life problem (e.g., Smart Farming, Personal Finance Tracker).	
	Inheritance + Method Overriding	- Parent/Child classes - super(), override methods	- Person → Student & Teacher classes - Each with a unique method	Mini Project 3: OOP Library System (borrow, return, show)		
Week 4	File Handling + Exceptions	- Read/Write .txt files - try/except/finally - Custom exceptions	- File-based diary app - Add exception handling	Comment-based exception flow documentation	Each group delivers a 3-minute presentation on any Python or AI tech trend (e.g., Generative AI, Pandas in Data Science).	
	Recap + Practice + Quiz	- Revise Python + OOP - Pattern printing - List/Dict Comprehension	- 5 real-world logic functions - Solve 3 HackerRank problems	Checkpoint: Quiz + Lab Task		

Complete this Apptitute Reasoning part with level to pass this level

Levels	Aptitude Details	Reasoning Details	IT Context	Resources
Level 1C				
1C	Time, Speed & Distance; Trains, Boats & Streams	Linear Sitting Arrangements	Data transfer, latency, UI arrangement	Distance calculators, UI design tasks
	Partnership: Profit sharing	Input-Output (Number/Word transformations)	Resource sharing, data processing	Flowchart tools, algorithms
	Pipes & Cisterns problems	Statement & Conclusion	Server load, bandwidth use, logs inference	Puzzle generators, logs
	Permutations & Combinations, Calendar & Clocks	Calendar dates, Clock angles	Passwords, config options, task scheduling	Calculators, Calendar puzzles

Level 1C				
Sr. No	**Topic**	**Task / Programs**	**Status**	**Category**
1	Arrays	Rotate array by k steps		DSA
2	Arrays	Move all zeros to end		DSA
3	Arrays	Find majority element (n/2 times)		DSA
4	Arrays	Two Sum problem		DSA
5	Arrays	Kadane's Algorithm (max subarray sum)		DSA
6	Arrays	Find subarray with given sum		DSA
7	Arrays	Find equilibrium index		DSA
8	Arrays	Count pairs with given sum		DSA
9	Arrays	Sort 0s, 1s, 2s (Dutch National Flag)		DSA
10	Arrays	Leaders in an array		DSA
11	Strings	Longest Palindromic Substring		DSA
12	Strings	Longest Repeating Character Replacement		DSA
13	Strings	Group Anagrams		DSA
14	Strings	Valid Parentheses		DSA
15	Strings	Minimum window substring		DSA
16	Strings	Substring with concatenation of all words		DSA
17	Strings	Longest substring without repeating characters		DSA
18	Strings	Count and Say problem		DSA
19	Strings	Rabin-Karp string matching		DSA
20	Strings	Z-Algorithm / KMP Pattern Searching		DSA
21	Linked List	Intersection point of two linked lists		DSA
22	Linked List	Add two numbers represented by linked lists		DSA
23	Linked List	Flatten a linked list		DSA
24	Linked List	Rotate linked list		DSA
25	Linked List	Check palindrome linked list		DSA
26	Linked List	Remove N-th node from end		DSA
27	Linked List	Odd-Even linked list		DSA
28	Stack & Queue	Min stack problem		DSA
29	Stack & Queue	Evaluate postfix expression		DSA
30	Stack & Queue	Infix to postfix conversion		DSA
31	Stack & Queue	Largest rectangle in histogram		DSA
32	Stack & Queue	Sliding window maximum using deque		DSA
33	Stack & Queue	Circular queue implementation		DSA
34	Stack & Queue	LRU Cache with deque + hashmap		DSA
35	Recurring & Backtracking	Rat in a maze		DSA
36	Recurring & Backtracking	N-Queens problem		DSA
37	Recurring & Backtracking	Sudoku solver		DSA
38	Recurring & Backtracking	Generate balanced parentheses		DSA
39	Recurring & Backtracking	Word search in grid		DSA
40	Recurring & Backtracking	Subset sum problem		DSA
41	Recurring & Backtracking	Print all subsets of a set		DSA
42	Searching & Sorting	Binary Search		DSA
43	Searching & Sorting	First and last occurrence of an element		DSA
44	Searching & Sorting	Count occurrences using binary search		DSA
45	Searching & Sorting	Find square root using binary search		DSA
46	Searching & Sorting	Search in rotated sorted array		DSA
47	Searching & Sorting	Median of two sorted arrays		DSA
48	Searching & Sorting	Find peak element		DSA
49	Searching & Sorting	Aggressive cows (binary search on answer)		DSA
50	Searching & Sorting	Book allocation problem		DSA
51	Searching & Sorting	Painter's partition problem		DSA
52	Searching & Sorting	QuickSort implementation		DSA
53	Trees & Binary Trees	Lowest common ancestor (LCA)		DSA
54	Trees & Binary Trees	Check if tree is balanced		DSA
55	Trees & Binary Trees	Boundary traversal		DSA
56	Trees & Binary Trees	Vertical order traversal		DSA

57	Trees & Binary Trees	Zigzag level order traversal		DSA
58	Trees & Binary Trees	Top view of binary tree		DSA
59	Trees & Binary Trees	Bottom view of binary tree		DSA
60	Trees & Binary Trees	Maximum path sum in binary tree		DSA
61	Trees & Binary Trees	Serialize and deserialize a tree		DSA
62	Trees & Binary Trees	Build tree from inorder & preorder		DSA
63	Graphs	Dijkstra’s algorithm		DSA
64	Graphs	Bellman-Ford algorithm		DSA
65	Graphs	Floyd-Warshall algorithm		DSA
66	Graphs	Prim’s MST algorithm		DSA
67	Graphs	Kruskal’s MST algorithm		DSA
68	Graphs	Clone graph		DSA
69	Graphs	Course schedule problem		DSA
70	Graphs	Bipartite graph check		DSA
71	Graphs	Word ladder problem		DSA
72	Graphs	Snake and ladder problem		DSA

Level 1C (Python)				
Sr. No	**Topic Name**	**Task / Programs**	**Status**	**Category**
1	Python Basics, Variables, Input/Output	1. Write a program to print Hello World. 2. Write a program to take user name as input and print a welcome message. 3. Write a program to input two numbers and print their sum. 4. Write a program to check the data type of user input. 5. Write a program to convert string input into integer. 6. Write a program to swap two variables. 7. Write a program to calculate simple interest. 8. Write a program to convert Celsius to Fahrenheit. 9. Write a program to take boolean input and print its opposite value. 10. Write a program to take multiple inputs in a single line.		
2	Conditional Statements (if, elif, else)	1. Write a program to check whether a number is even or odd. 2. Write a program to check whether a number is positive, negative, or zero. 3. Write a program to find the greatest of two numbers. 4. Write a program to find the greatest of three numbers. 5. Write a program to check whether a year is a leap year. 6. Write a program to check whether a number is divisible by 5 and 11. 7. Write a program to check pass or fail using marks. 8. Write a program to display grade using marks. 9. Write a program to check whether a character is a vowel or consonant. 10. Write a program to create a simple calculator using if-elif.		
3	Loops (for, while, range)	1. Write a program to print numbers from 1 to 10. 2. Write a program to print even numbers between 1 and 50. 3. Write a program to print odd numbers using while loop. 4. Write a program to print multiplication table of a number. 5. Write a program to find the sum of first n natural numbers. 6. Write a program to find factorial of a number. 7. Write a program to reverse a number. 8. Write a program to count digits in a number. 9. Write a program to print Fibonacci series. 10. Write a program to check whether a number is prime. 11. Write a program to print numbers using range(). 12. Write a program to find the sum of digits of a number. 13. Write a program to print square of numbers from 1 to n. 14. Write a program to find power of a number. 15. Write a program to count vowels in a string.		
4	break, continue	1. Write a program to stop loop execution when number equals 5. 2. Write a program to skip even numbers using continue. 3. Write a program to exit loop when negative number is entered. 4. Write a program to find the first multiple of 7 using break. 5. Write a password validation program using break.		
5	Functions (def, return, parameters)	1. Write a function to add two numbers. 2. Write a function to check whether a number is even or odd. 3. Write a function to find factorial of a number. 4. Write a function to check whether a number is prime. 5. Write a function to return the maximum of two numbers. 6. Write a function to reverse a string. 7. Write a function with default parameters. 8. Write a function to count vowels in a string. 9. Write a function to calculate area of a circle. 10. Write a function that returns multiple values.		
6	Lists (slicing, methods)	1. Write a program to create and print a list. 2. Write a program to add elements to a list. 3. Write a program to remove elements from a list. 4. Write a program to find length of a list. 5. Write a program to find maximum and minimum element in a list. 6. Write a program to slice first 3 elements from a list. 7. Write a program to reverse a list. 8. Write a program to sort a list. 9. Write a program to count occurrences of an element in a list. 10. Write a program to merge two lists.		
7	Tuples	1. Write a program to create and access a tuple. 2. Write a program to convert tuple into list. 3. Write a program to find index of an element in tuple. 4. Write a program to count occurrences in tuple. 5. Write a program to swap two variables using tuple.		
8	Dictionary (CRUD Operations)	1. Write a program to create a dictionary. 2. Write a program to access values from dictionary. 3. Write a program to add a new key-value pair in dictionary. 4. Write a program to update dictionary value. 5. Write a program to delete a key from dictionary. 6. Write a program to iterate dictionary using loop. 7. Write a program to count frequency of characters using dictionary.		
9	Sets	1. Write a program to create a set and add elements. 2. Write a program to find union of two sets. 3. Write a program to find intersection of two sets. 4. Write a program to find difference of two sets.		
10	Strings	1. Write a program to convert string into lowercase. 2. Write a program to replace a word in string. 3. Write a program to count number of words in string. 4. Write a program to reverse a string. 5. Write a program to check palindrome string. 6. Write a program to find substring in string.		
11	OOP – Classes & Objects	1. Write a program to create a class with attributes. 2. Write a program using init method. 3. Write a program to create object and access attributes. 4. Write a program to create method inside a class. 5. Write a program for Student class. 6. Write a program for Bank Account class.		
12	Inheritance & Polymorphism	1. Write a program demonstrating single inheritance. 2. Write a program using super() keyword. 3. Write a program to override a method. 4. Write a program demonstrating multiple inheritance.		
13	File Handling & Exception Handling	1. Write a program to create and write data into a text file. 2. Write a program to read data from a text file. 3. Write a program to append data to a file. 4. Write a program to count number of lines in a file. 5. Write a program to handle division by zero exception. 6. Write a program using try-except-finally. 7. Write a program to create a custom exception.		
14	Comprehension & Advanced	1. Write a program using list and dictionary comprehension.		